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Indium-copper-rich sphalerite from the Restauradora vein, Capillitas, Catamarca, Argentina

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Abstract

The Restauradora vein, is one of the 20 veins of Capillitas (27°27' S, 66°30' W), an epithermal precious- and base-metal vein deposit belonging to the Farallón Negro Mining District, northwestern Argentina. One of the main minerals of the paragenesis of Restauradora vein is sphalerite, showing a large substitution of Zn by In and Cu with small participation of Cd and Fe; the fluctuating Zn composition ranges from 0.99 to 0.49 *apfu*, while In and Cu ranges (*apfu*) are: 0.22–0.27 and 0.21–0.25, respectively. Under the polarizing-light microscope, using reflected light, the optical properties are those of sphalerite, but with decreasing contents of Zn, especially when Zn is ≤ 0.60 *apfu*, a new variety differentiates from regular sphalerite and develops ribbons with paler colors, higher reflectance, softer relative hardness and no internal reflections. The main reflectance values of this In-Cu-rich sphalerite are (*R*_{air}, λ /nm): 20.55, 470; 19.43, 546; 19.02, 589; and 18.42, 650. The strongest five x-ray powder-diffraction lines of the synthetic analogue [*d* in Å (*I*) (*hkl*)] are: 3.173 (100) (111); 1.943 (54) (202); 1.657 (37) (113); 1.122 (22) (224); 1.261 (19) (133), the space group is *F*-43*m*, with *a* = 5.49692(10) Å, *V* = 166,100(10) Å³, *Z* = 4. Electron-microprobe analyses of the In-Cu-rich sphalerite grains (*n* = 10) give an average composition (wt%) of: S 29.51, Zn 31.03, In 24.89, Cu 13.49, Ga 0.96, Cd 0.54, Fe 0.05; the average (*n* = 7) chemical composition of the synthetic phase is: S 29.67, Zn 30.32, In 25.94, Cu 13.82. Despite of the differences, and given that Zn is the main cation and that the structure is the same as that of sphalerite, the mineral studied here is considered as the richest In-Cu variety of sphalerite in the world.

KEYWORDS

Argentina, Capillitas, epithermal, Farallón Negro Mining District, indium, sphalerite

1 | INTRODUCTION

While studying the minerals of Restauradora vein (Márquez-Zavalía et al., 2020; Márquez-Zavalía & Pedregosa, 1994;

Sejkora et al., 2022), Capillitas, Argentina, it was observed a ribbon-like mineral of lighter color than sphalerite, associated to that mineral. The ribbons are generally thinner than a few microns, reaching in occasions up to 15 µm. A deeper

research was going to show if the mineral from Restauradora vein was a new species or a In-Cu-rich variety of sphalerite, since this mineral is well known to accommodate a variety of minor and trace elements (e.g., Fe, Cu, In, Ga, Ge) in its structure (e.g., Bauer et al., 2019; Cook et al., 2009; Filimonova et al., 2019; Goldmann et al., 2019; Mederski et al., 2022; Murakami & Ishihara, 2013; Sahlström, Arribas, et al., 2017; Sahlström, Blake, et al., 2017; Shimizu & Morishita, 2012; Voudouris et al., 2022). The preliminary results and their likeness to two described unknown minerals in similar paragenesis, one at Toyoha mine, Japan (Ohta, 1980, 1989) and one at the Pravouriiskoe deposit, Russia (Semenyak et al., 1994), encourage us to perform that deeper study on the subject and the results are shown in this paper.

2 | OCCURRENCE

Restauradora vein is one of the 20 veins of Capillitas, an epithermal precious- and base-metal vein deposit (Márquez-Zavalía, 1988, 1999; Putz et al., 2009), located

along the eastern slope of the Capillitas Range, in Catamarca province, northwestern Argentina (Figure 1). Capillitas is genetically linked to other deposits of the Farallón Negro Mining District (Márquez-Zavalía & Heinrich, 2016 and references therein), such as the world-class porphyry deposits Bajo de la Alumbrera and Agua Rica and the smaller El Durazno, three other epithermal deposits: Farallón Negro, Alto de la Blenda and Cerro Atajo, and several smaller Au-bearing porphyry prospects with extensive alteration including Agua Tapada, El Espanto, San Lucas and Filo Colorado.

In the Capillitas area, the rocks of the Farallón Negro Volcanic Complex form an ellipsoidal diatreme (1500 m × 900 m); hydrothermal alteration is widespread, varying laterally by host-rock type. The veins are hosted by the volcanic and granitic rocks in ENE and WNW directions (Márquez-Zavalía, 1999) and the mineralization is very diverse with more than 120 minerals among primary and secondary species. Capillitas is the type locality of five minerals: putzite (Paar et al., 2004), catamarcaite (Putz et al., 2006), isiharaite (Márquez-Zavalía et al., 2014),

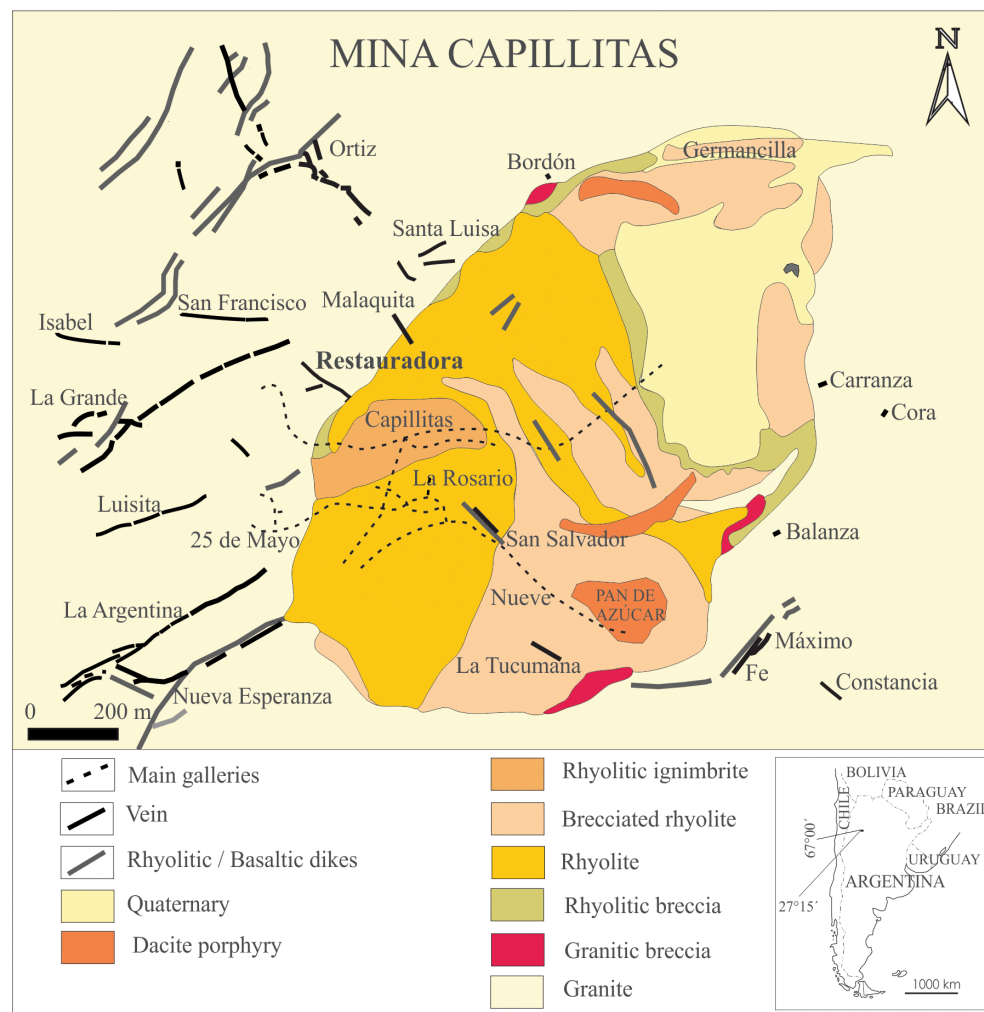


FIGURE 1 Schematic geology of Mina Capillitas (modified from Márquez-Zavalía & Craig, 2004).

liskirchnerite (Effenberger et al., 2015), and omariniite (Bindi et al., 2017). The deposit has been discontinuously mined for gold since pre-Colombian times. Over the last two centuries extensive work was carried out with sporadic copper recovery attempts, interrupted by metallurgical problems mainly due to insufficient mineralogical knowledge of the ore to be treated. Since the second part of the last century, mining has been almost exclusively dedicated to the extraction of banded—and sporadically stalactitic—rhodochrosite, the main gangue mineral in some of the veins (Márquez-Zavalía & Craig, 2022 and references therein).

The Restauradora vein, hosted in granites at 3290 m o. s.l., has an average thickness of 50 cm, and a total measured length of 106 m. The vein has two branches with strike and dip N10° W/N70° E and 75° E–75° W/70° S, respectively. The average grades reported for this vein (JICA, 1978–1981) are: 0.6 g/t Au, 209 g/t Ag, 7.10% Cu, 1.11% Pb, 1.49% Zn, and 0.07% Mn. The works performed along this vein consist of a 100 m long gallery, two inclines and a shaft. There are two dumps, one over the gallery level and the other below it, formed from the droppings of the one above; most of the material is oxidized.

Only two of the seven stages of mineralization recognized in the Capillitas deposit (Márquez-Zavalía, 1988; Márquez-Zavalía & Craig, 2004) are represented in the Restauradora vein and the In-Cu-rich sphalerite is present in the later stage (Figure 2). The main minerals found at this vein are: pyrite, sphalerite s.s. and its In-Cu-rich variety, chalcopyrite, tennantite-(Zn), tennantite-(Fe), galena, and enargite. The accessory and secondary ore minerals, sometimes very scarce, are: bornite, stannite, famatinite, gold

and digenite, chalcocite, covellite and goethite, accompanied by local transparent species represented by antlerite, cyanotrichite, carbonate-cyanotrichite, grandviewite, linarite, malachite, and gypsum; the main gangue mineral is quartz.

3 | ANALYTICAL METHODS

Fieldwork at Capillitas was conducted during several field trips; samples were obtained from all the veins of the deposit either in the underground workings, or at outcrops and dumps.

Microscopic examinations of ore minerals were made on polished thin sections and standard polished sections under a reflected-light polarizing microscope, before carbon-coating for electron-microprobe analysis.

The small grain size of In-Cu-rich sphalerite, and the lack of single crystals, prevented its extraction and isolation in an amount required for relevant crystallographic and structural investigations, hence, a synthetic analogue was made. The synthetic phase was prepared using an evacuated silica glass tube method in the Laboratory of Experimental Mineralogy of the Czech Geological Survey in Prague. High purity sulfur, zinc, indium, and copper were used as starting materials for the synthesis. The evacuated tube with its charge was sealed and annealed at 300°C for 25 days. After cooling by cold-water bath, the charge was ground into powder in acetone using an agate mortar, and thoroughly mixed to homogenize. The pulverized charge was sealed in an evacuated silica-glass tube again, and reheated at 200°C for 3 months. The experimental product was rapidly quenched in cold water.

Electron back-scattering diffraction (EBSD) technique was used for a crystallographic characterization of the natural sample. The main objective of the performed EBSD measurement was to investigate, if the natural sample shows disordered cubic sphalerite structure or an ordered one. A TESCAN Mira 3GMU scanning electron microscope combined with EBSD system (NordlysNano detector, Oxford Instruments) was used for the measurements. The natural sample was prepared for EBSD investigation by etching the mechanically polished surface with colloidal silica (OP-U) for 30 min to reduce the surface damage. The EBSD patterns were collected and processed using a proprietary computer program AZtec HKL (Oxford Instruments).

X-ray powder diffraction data were collected on the synthetic analogue of the In-Cu-rich sphalerite, in Bragg–Brentano geometry on a Bruker D8 Advance powder diffractometer. The CuK α radiation, and a Lynx Eye-XE detector were used. The data were collected in the angular range from 10 to 120° of 2 θ with a 0.015° step.

Minerals	Hypogene ore minerals		Supergene ore minerals
	Stage III	Stage IV	
Pyrite	—	—	
Sphalerite	—	—	
Galena	—	—	
Chalcopyrite	—	—	
Enargite	—	—	
Tennantite-(Zn)	—	—	
Tennantite-(Fe)	—	—	
<i>In-Cu-rich sphalerite</i>		—	
Gold ($\pm$ Ag)		—	
Bornite		—	
Stannite		—	
Famatinite		—	
Digenite			—
Chalcocite			—
Covellite			—
Goethite			—

FIGURE 2 Schematic paragenetic sequence of Restauradora vein.

Chemical analyses of the natural sample were performed with a Cameca SX-50 electron microprobe at the Department of Geological Sciences, University of Manitoba, Winnipeg, Canada, and with a JEOL JXA-8900R, at the Geological Survey of Japan, Tsukuba, Japan. Accelerating voltage and beam current were kept at 15 kV, 12 nA on the Faraday cup, respectively and the electron

beam diameter was 2 μm , the standards and analytical lines selected were: S, Fe, Cu (CuFeS₂, K α), Zn (ZnS, K α), Ga (GaAs, L α), Cd (CdS, L α), In (InP, L α), Ge (Ge, L α), As (InAs, L α) and Sb (Sb₂Te₃, L α). Manganese, Co, Ni, Sn, Te, Pb, and Bi were also sought for but all results were below detection limits; Ge, As, and Sb were above limits of detection in some analyses. The ZAF method was employed for matrix corrections.

The synthetic material was analyzed using a Cameca SX-100, WDS mode, 15 kV, 20 nA, 1–2 μm beam diameter at the Institute of Geology of the Czech Academy of Sciences. The standards and analytical lines selected were as follows: S (ZnS, K α), Cu (Cu, K α), Zn (Zn, K α) and In (InP, L α). Data were reduced using the PAP routine of Pouchou and Pichoir (1985).

The reflectance values were determined using a Leitz Orthoplan microscope, attached to a Hamamatsu solid state mini-spectrometer connected by optical fiber, with dry and immersion objectives 20X, (effective NA: 0.2 in air) using SiC No. 878 as standard.

TABLE 1 In-Cu-rich sphalerite and sphalerite reflectance values in air.

In-Cu-rich sphalerite ^a		Sphalerite ^b	
λ (nm)	R_{air}	λ (nm)	R_{air}
400	20.96	400	19.6
420	20.95	420	19.0
440	20.86	440	18.3
460	20.65	460	17.9
470 (COM)	20.55		
480	20.40	480	17.5
500	20.10	500	17.2
520	19.80	520	16.9
540	19.51	540	16.7
546 (COM)	19.43		
560	19.34	560	16.5
580	19.19	580	16.4
589 (COM)	19.02		
600	18.97	600	16.3
620	18.71	620	16.2
640	18.51	640	16.1
650 (COM)	18.42		
660	18.36	660	16.0
680	18.25	680	15.9
700	18.11	700	15.8

Note: Bold values correspond to the COM standard values required.

^aThis paper.

^bQDF III by Criddle and Stanley: Sphalerite 523, (in Anthony et al. 2003: <http://www.handbookofmineralogy.org/>).

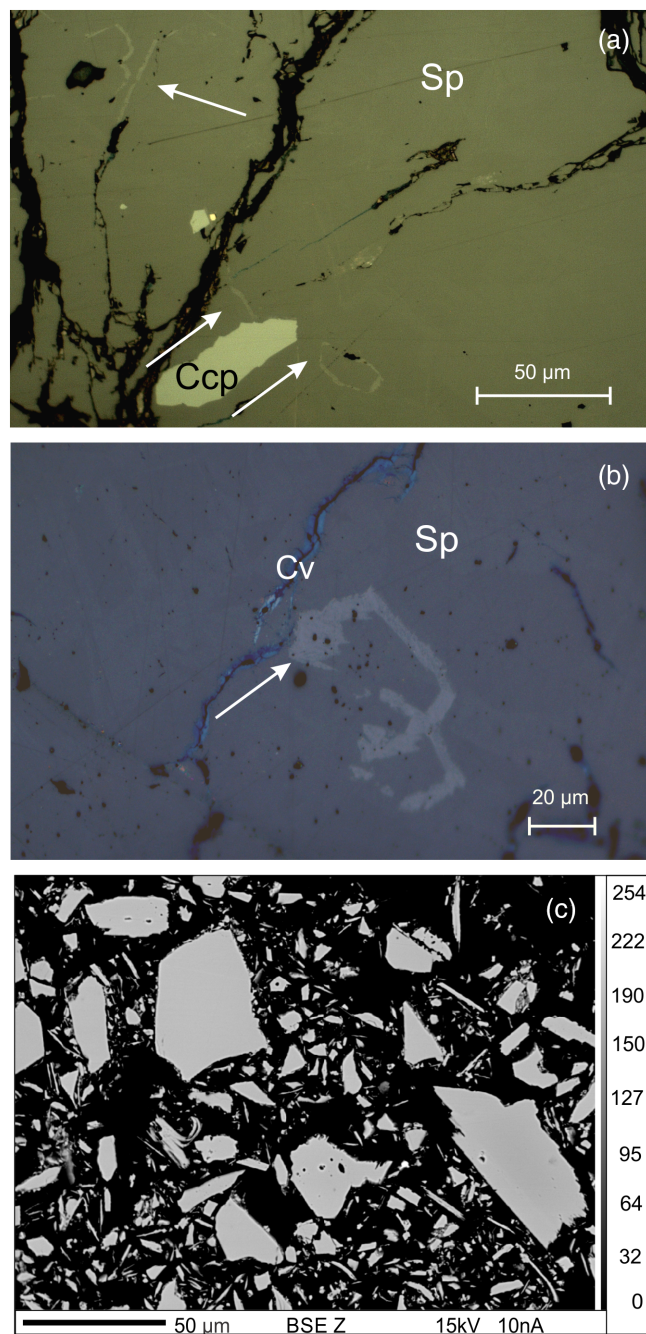


FIGURE 3 (a, b) Photomicrography of section RE-752. In-Cu-rich sphalerite surrounded by regular sphalerite (Sp) associated with chalcopyrite (Ccp), thin veinlet replaced by covellite (Cv); (c) BSE image of the synthetic sample. Arrows point to In-Cu-rich sphalerite.

4 | IN-CU-RICH SPHALERITE

4.1 | Physical properties

Under the reflected light microscope, the In-Cu-rich sphalerite is opaque and occurs as thin, and generally discontinuous, ribbons usually up to 10–15 μm thick and up to 50 μm long (Figure 3a,b—in oil and air, respectively). Its color is olive gray in air and in oil, it is softer and has higher reflectance than the host mineral, which is always regular sphalerite. Neither birrefraction nor pleochroism were observed; in none of the sections internal reflections were perceived despite careful observation.

TABLE 2 X-ray powder diffraction data (d in Å) of the synthetic sample.

I_{meas}	d_{meas} Å	d_{calc} Å	hkl
100	3.1729	3.1735	111
10	2.7483	2.7483	2
54	1.9432	1.9433	202
37	1.6573	1.6573	113
7	1.5868	1.5867	222
8	1.3743	1.3742	4
19	1.261	1.261	133
4	1.2291	1.2291	402
22	1.122	1.122	224
18	1.0579	1.0578	511/333
7	0.9717	0.9717	44
16	0.9291	0.9291	135
1	0.9161	0.9161	6
10	0.8691	0.8691	206
11	0.8382	0.8382	533

Note: Bold values correspond to the strongest five X-ray powder-diffraction lines.

About 700 reflectance values for one spectrum were obtained, subsequently reduced to the needed wavelength 400 to 700 nm in steps of 20 nm, plus the 4 COM standard wavelengths. The obtained values (R_1 air) are shown in Table 1, compared to those for sphalerite (Criddle & Stanley: Sphalerite 523, in Anthony et al., 2003: <https://www.handbookofmineralogy.org/pdfs/sphalerite.pdf>).

4.2 | XRD crystallography

A synthetic analogue of In-Cu-rich sphalerite was prepared to compensate for the lack of suitable material for relevant crystallographic and structural investigations. The powder-crystal x-ray diffraction data of the synthetic phase are shown in Table 2.

The crystal structure of the synthetic sample (Figure 3c) was refined using the Rietveld method by the Topas 5 program (Bruker AXS, 2014). The crystal sphalerite structure (Hall, 1961) was used as an initial structure model in the Rietveld refinement. All metals (Cu, In and Zn) occupy the same crystallographic site (i.e., the $4a$ position of the $F\bar{4}3m$ space group); their occupancy factors were set to correspond the chemical composition ($\text{Zn}_{0.5}\text{In}_{0.25}\text{Cu}_{0.25}\text{S}_{1.00}$ (for $Z = 4$)). The Rietveld refinement involved refinement of unit-cell parameter, isotropic size and isotropic displacement parameters. As all atom positions are in special sites, no crystallographic coordinates were refined. In total, 5 parameters were refined; background was determined by means of a Chebychev polynomial of the 5th order. The final Rietveld plot is shown in Figure 4.

The EBSD patterns (also known as Kikuchi patterns) obtained from the natural material (27 measurements on different spots) were found to match the patterns generated from the refined structural model for the synthetic

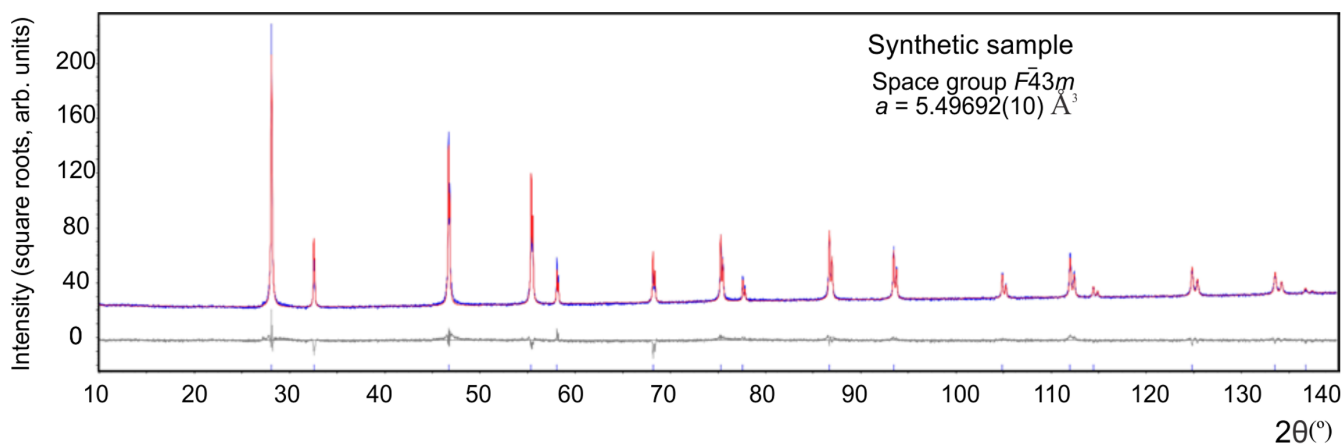


FIGURE 4 The final Rietveld plot for the synthetic sample.

phase (Figure 5) and confirmed a cubic sphalerite structure for the natural phase. The values of the mean angular deviation (MAD, i.e., goodness of fit of the solution) between the calculated and measured Kikuchi bands

range between 0.21 and 0.36°. These values reveal a very good match; as long as values of mean angular deviation are less than 1°, they are considered as indicators of an acceptable fit (HKL Technology, 2004).

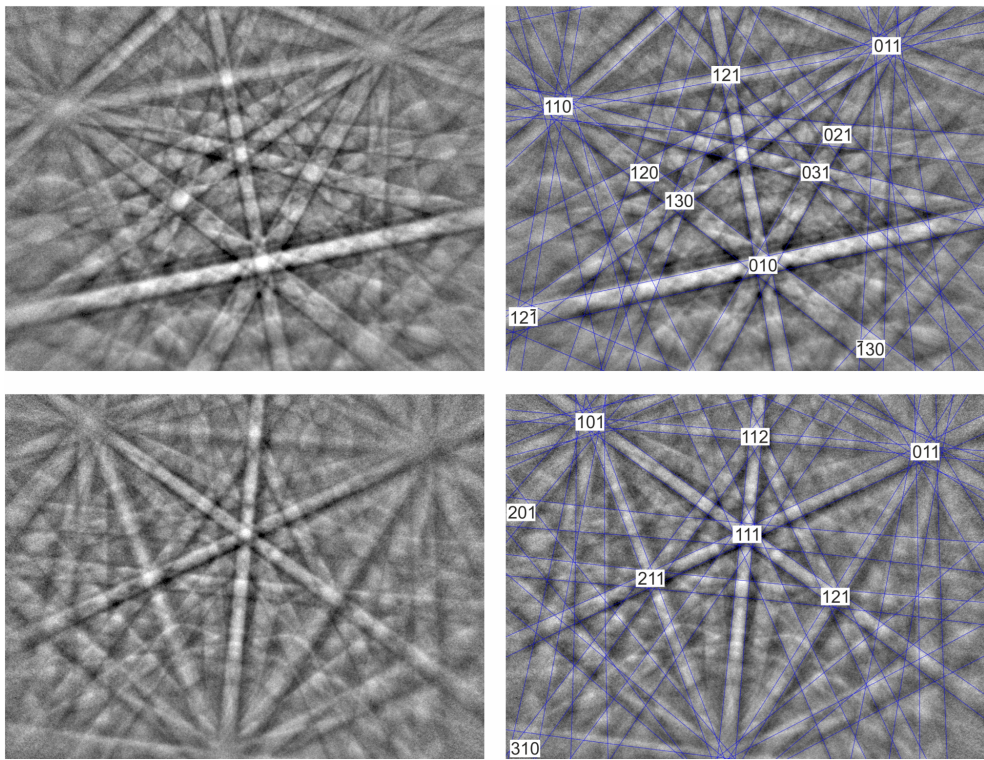


FIGURE 5 EBSD images of In-Cu-rich sphalerite at different orientation of grains; in the right pane the Kikuchi bands are indexed.

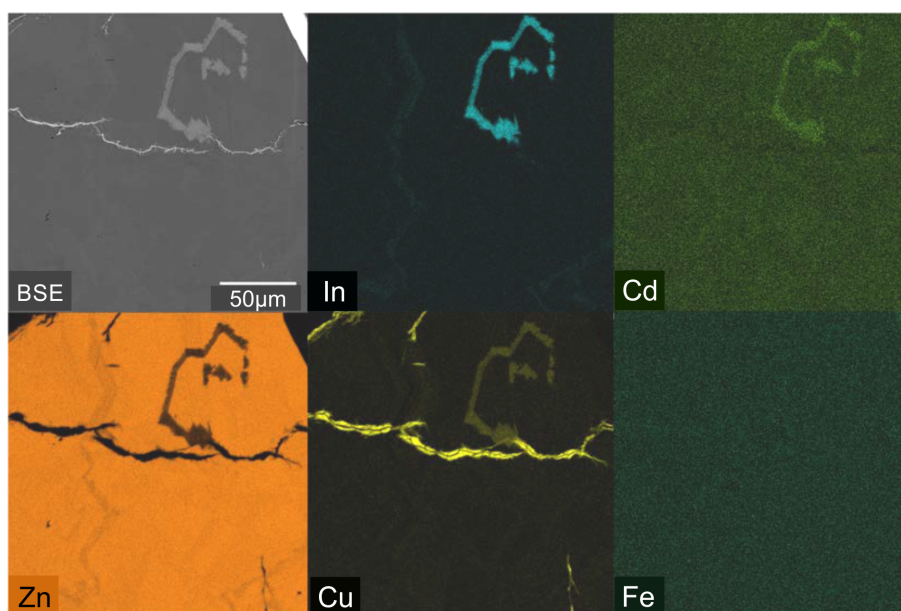
TABLE 3 Representative chemical compositions of the In-Cu-rich sphalerite.

No.	UK-01	UK-02	UK-03	UK-04	UK-05	UK-06	UK-07	UK-08	UK-09	UK-10	AVG	SD
S wt%	29.21	30.03	29.73	29.49	29.82	28.84	30.01	29.69	29.66	28.64	29.51	0.45
Zn	34.19	32.75	31.67	31.74	32.10	31.21	29.26	29.27	29.70	28.42	31.03	1.73
In	22.83	23.26	24.44	24.46	24.72	25.11	25.34	25.37	26.54	26.88	24.89	1.20
Cu	12.34	12.56	13.76	12.47	13.50	14.06	14.02	13.64	14.16	14.37	13.49	0.72
Cd	0.61	0.44	n.a.	n.a.	0.68	n.a.	0.44	0.56	0.52	n.a.	0.54	0.09
Ga	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	0.96	0.96	0.00
Fe	0.04	n.a.	0.01	0.11	n.a.	0.03	n.a.	n.a.	n.a.	n.a.	0.05	0.04
Total	99.22	99.04	99.61	98.27	100.82	99.25	99.06	98.53	100.58	99.27	99.51	
S apfu	0.998	1.021	1.011	1.017	1.007	0.994	1.027	1.025	1.010	0.995	1.011	0.01
Zn	0.564	0.538	0.520	0.529	0.524	0.519	0.484	0.488	0.489	0.477	0.513	0.03
In	0.218	0.221	0.232	0.236	0.233	0.242	0.242	0.245	0.252	0.261	0.238	0.01
Cu	0.213	0.216	0.236	0.217	0.230	0.244	0.242	0.237	0.243	0.252	0.233	0.01
Cd	0.006	0.004	0.000	0.000	0.007	0.000	0.004	0.006	0.005	0.000	0.003	0.00
Ga	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.015	0.002	0.00
Fe	0.001	0.000	0.000	0.002	0.000	0.001	0.000	0.000	0.000	0.000	0.000	0.00
Total		2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000		

Note: Mn, Co, Ni, Ge, As, Sn, Sb, Te, Pb, and Bi below detection limits.

TABLE 4 Representative chemical compositions of the synthetic analogue of In-Cu-rich sphalerite.

Analysis #	Syn-01	Syn-02	Syn-03	Syn-04	Syn-05	Syn-06	Syn-07	Avr.	SD
S wt%	29.54	29.71	29.26	29.90	29.59	30.33	29.38	29.67	0.33
Zn	30.63	32.03	28.48	31.83	28.92	31.51	28.85	30.32	1.42
In	26.00	24.93	27.02	24.65	26.96	25.43	26.61	25.94	0.89
Cu	14.22	13.62	14.16	12.94	14.19	13.72	13.88	13.82	0.42
Total	100.39	100.29	98.92	99.31	99.66	100.99	98.72	99.75	
S <i>apfu</i>	1.059	1.058	1.069	1.068	1.071	1.070	1.072	1.067	0.01
Zn	0.538	0.559	0.511	0.557	0.513	0.545	0.516	0.534	0.02
In	0.260	0.248	0.276	0.246	0.272	0.250	0.271	0.261	0.01
Cu	0.142	0.135	0.145	0.129	0.143	0.135	0.141	0.139	0.01
Total	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000	

FIGURE 6 BSE image and WDS mapping of the section RE-752.

4.3 | Chemical composition

The chemical composition (Table 3) of this In-Cu-rich sphalerite (range [average ($n = 10$), SD (2σ)] in wt%) is: S: 28.64–30.03 [29.51, 0.45], Zn: 28.42–34.19 [31.03, 1.73], In: 22.83–26.88 [24.89, 1.20], Cu: 12.34–14.37 [13.49, 0.72], Cd: 0.44–0.68 [0.54, 0.09] Fe: 0.01–0.11 [0.05, 0.04]; one analysis gave 0.96 wt% Ga. The empirical formula calculated from the average of the 10 chemical analyses given in Table 3, and based on 2 *apfu*, is: $(\text{Zn}_{0.513}\text{In}_{0.238}\text{Cu}_{0.233}\text{Cd}_{0.003}\text{Ga}_{0.002})\Sigma_{0.989}\text{S}_{1.011}$.

The chemical composition of the In-Cu-rich sphalerite synthetic analogue (range [average ($n = 7$), SD (2σ)] in wt%) is: S: 29.26–30.33 [29.67, 0.33], Zn: 28.48–32.03 [30.32, 1.42], In: 24.65–27.02 [25.94, 0.89], Cu: 12.94–14.22 [13.82, 0.42]. The empirical formula calculated

from the 7 chemical analyses given in Table 4, and based on 2 *apfu*, is: $(\text{Zn}_{0.534}\text{In}_{0.261}\text{Cu}_{0.139})\Sigma_{0.933}\text{S}_{1.067}$.

Figure 6 shows the BSE image and the elemental distribution of the In, Cd, Zn, Cu, and Fe of the In-Cu-rich sphalerite from one of the polished sections (RE-752).

5 | DISCUSSION

After carrying out the studies on the ribbon-like mineral, we concluded that it is a variety of sphalerite extremely rich in In and Cu.

The In-Cu-rich sphalerite was formed in the latest mineralization stage (Figure 2: Stage IV) of the Restauradora vein, while during the earlier stage (Figure 2: Stage III) precipitated sphalerite without In or Cu in its composition.

The microthermometric data obtained from the two-phases fluid (L + V) hosted in the quartz of the two stages of the mineralization represented in this vein gave an average homogenization temperature of 250°C and a salinity of 3 wt% NaCl_{equiv} (Márquez-Zavalía, 1988; Márquez-Zavalía & Craig, 2004). During the earlier stage of the mineralization, quartz, pyrite and sphalerite are accompanied by the Cu-rich minerals and galena, while on the later stage, the previously mentioned minerals continued depositing along with Au and small amounts of tennantite-(Fe), bornite, stannite, farnatite and the In-Cu-rich sphalerite (Figure 2). Similarly to what was described by Sinclair et al. (2006), the In-Cu-rich sphalerite in this vein, devoid of chalcopyrite inclusions, has higher contents of In, and consequently of Cu, when in the vicinities there are thin veinlets or small crystals of chalcopyrite, which are consistently free of In. The fact that chalcopyrite is generally free of indium in its structure when sphalerite is abundant was also noticed by Cook et al. (2011).

It is important to emphasize that some sphalerite grains associated with the variety rich in indium and copper do not contain these elements in their composition, while in other grains these elements are present in different and variable proportions. The In and Cu contents vary greatly between bands and tend to decrease toward the edge of the crystals (Márquez-Zavalía et al., 2020). Based on that and that there is no indication of cracks with higher In concentrations, we assumed that the In and Cu do not postdate the formation of sphalerite but they were concentrated in the hydrothermal fluids of the later stage of the mineralization.

When the indium reaches over 22 wt%, which is equivalent to 0.20 *apfu*, the ribbons of In-Cu-rich sphalerite are visible under the reflected light microscope (Table 3), while the grains with In < 0.20 *apfu* correspond to optically homogeneous sphalerite, and only the BSE images reveal the chemical zonation.

According to Schwarz-Schampera and Herzig (2002), we suggest that the physical-chemical conditions necessary

for the substitution of Zn by In-Cu agree with the prevailing environment of formation of the In-Cu-rich sphalerite from Capillitas, indicated by the homogenization temperatures and salinities obtained from fluid inclusions measured in quartz crystallized during the fourth stage of the mineralization and by the paragenetic minerals associated to it. High negative correlations (Table 5) are observed for the pairs Zn-In (Figure 7a) and Zn-Cu (Figure 7b), while the pair Zn-Ga (Figure 7c) shows a weak negative correlation. The pair In-Cu (Figure 7d) shows high positive correlation, while the pair Ga-Sb shows also high positive correlation (Table 5), but due to the few analyses where those elements coexist, the graphic does not correlate as well; the pair In-Ga (Figure 7e) shows positive correlation, but not as noticeable as the pair In-Cu. The pairs Zn-Cd, Zn-Ge, Zn-Fe, Zn-As, and Zn-Sb do not show significant, either positive or negative, correlations (Figure 7f). The most significant of the correlations corresponds to the pair In-Cu that in all analyses (Table 3) the ratio is consistently close to 1:1, then we agree with Johan (1988), Cook et al. (2011), Frenzel et al. (2016), and Sahlström, Blake, et al. (2017); Sahlström, Arribas, et al. (2017) that Cu⁺ is the primary cation involved in the linked substitution $2\text{Zn}^{2+} \leftrightarrow \text{Cu}^{+} + \text{In}^{+3}$.

As evidenced by the presence in the Capillitas veins of both high and low *f*Te₂ mineral phases (like calaverite and silver, respectively) it was assumed that the fluids experienced significant compositional fluctuation, possibly pulses, during the fourth stage of the mineralization (Márquez-Zavalía & Craig, 2004). As a result, and as it has already been postulated (e.g., Patrick et al., 1993; Sibson, 1981), zoning in sphalerite might be attributed to changes in the physical chemistry of the mineralizing fluids with local periodic increases on In and Cu that are related with recurrent reactivation of fractures or periodic inputs of meteoric water into the system. During the cooling and dilution of the hydrothermal fluids, the metals, many of them—including In—incompatible with

	Zn	In	Cu	Cd	Ga	Ge	Fe	As
Zn								
In	−0.98							
Cu	−0.94	0.95						
Cd	−0.27	0.31	0.39					
Ga	0.29	−0.31	−0.32	−0.64				
Ge	0.12	−0.12	−0.18	−0.14	0.08			
Fe	0.05	−0.06	−0.01	−0.27	0.32	0.06		
As	0.38	−0.44	−0.36	−0.29	0.41	0.05	0.13	
Sb	0.29	−0.31	−0.33	−0.65	0.89	0.15	0.37	0.30

TABLE 5 Correlation matrix for major and minor elements of In-Cu-rich sphalerite.

Note: The green colors in their different shades correspond to positive correlations, while the red-orange-yellow colors correspond to negative correlations, in descending order, respectively.

FIGURE 7 Binary plots showing the compositions of In-Cu-rich sphalerite.

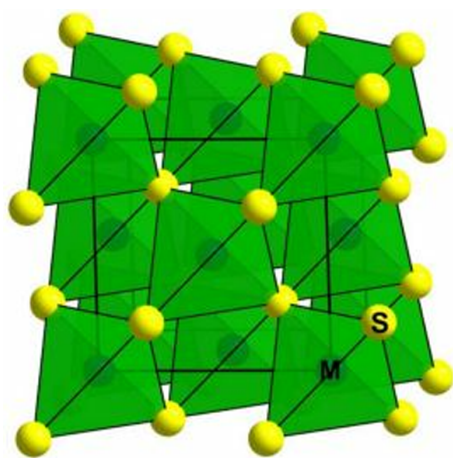
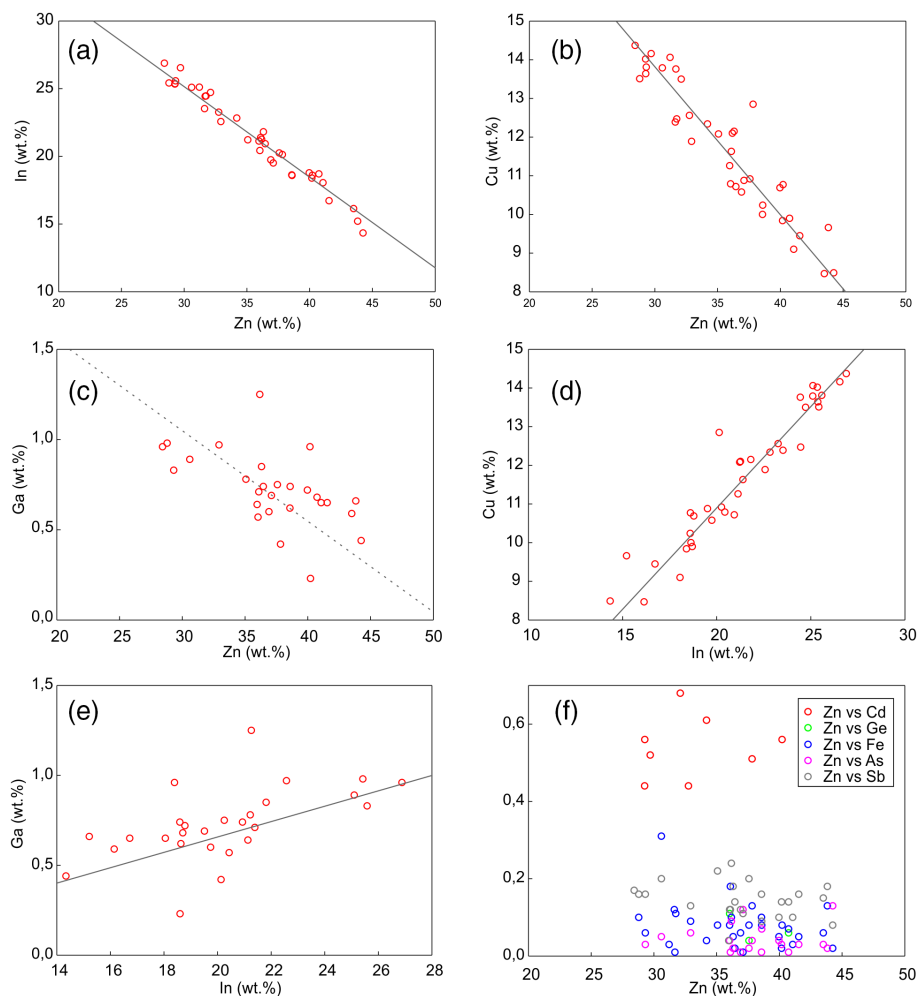


FIGURE 8 Polyhedral representation of the In-Cu-rich sphalerite crystal structure showing the corner-sharing $[MS_4]$ tetrahedra ($M = Cu, In, \text{ and } Zn$). The unit-cell edges are outlined.

magmatic crystallization and therefore concentrated in these fluids, were deposited. Similar mechanisms have been proposed (e.g., Sinclair et al., 2006 and referenced therein) for other indium-bearing polymetallic deposits.

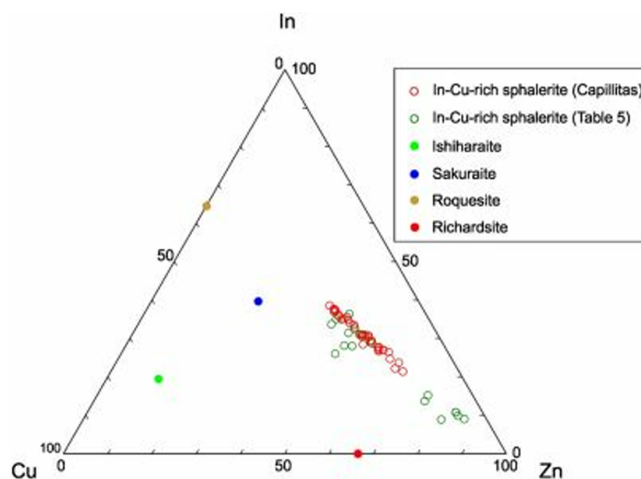


FIGURE 9 Ternary plot of Zn-In-Cu of In-Cu-rich sphalerite from Nueva Esperanza and Restauradora veins, compared *t* In-rich sphalerite from other localities, ishiharaite, sakuraiite, roquesite, and richardsite.

A low- to intermediate-salinity fluid that originated from a porphyry intrusion associated with the Farallón Negro Volcanic Complex may be the source of indium and other

TABLE 6 Representative chemical analyses of In-Cu-rich sphalerite from other localities.

Analysis	References	Fe wt%	Cu	Ag	Sn	In	Zn	Ga	Cd	S	Total	Fe apfu	Cu	Ag	Sn	In	Zn	Ga	Cd	S
2-6_10a-In35	Bauer et al. (2019)	3.42	11.11	–	–	20.14	33.92	–	0.35	29.81	98.75	0.066	0.188	0.000	0.000	0.188	0.557	0.000	0.000	0.998
7	Burke and Kieft (1980)	0.7	6.9	–	–	10.4	51.1	–	0.4	31.0	100.5	0.01	0.11	0.00	0.00	0.10	0.80	0.00	0.004	0.98
23	Dill et al. (2013)	0.49	4.10	0.02	0.40	7.31	56.72	0.01	0.52	31.51	101.15	0.009	0.065	0.000	0.003	0.064	0.870	0.000	0.005	0.985
j	Kieft and Damman (1990)	7.46	15.22	0.79	–	15.43	28.68	–	1.99	30.00	99.57	0.140	0.251	0.008	0.000	0.141	0.460	0.000	0.019	0.981
MG11CI	Moura et al. (2007)	4.77	3.79	–	–	6.73	51.97	–	0.51	31.35	99.12	0.09	0.06	0.00	0.00	0.06	0.80	0.00	0.00	0.99
13	Murao and Furuno (1990)	6.26	12.43	0.00	0.48	19.60	30.38	–	0.00	30.27	99.42	0.119	0.207	0.000	0.004	0.181	0.491	0.000	0.000	0.998
H1	Murakami and Ishihara (2013)	12.79	8.81	0.95	0.00	15.86	30.62	–	0.24	31.50	100.77	0.238	0.145	0.013	0.000	0.145	0.473	0.000	0.000	0.995
H2	Murakami and Ishihara (2013) ^a	12.77	10.08	0.97	0.00	18.09 ^a	27.69	–	0.22	31.16	82.89	0.238	0.165	0.013	0.000	0.165	0.433	0.000	0.000	0.990
SR350	Ohta (1989)	2.71	7.51	0.16	1.74	8.86	48.40	–	–	31.09	100.47	0.05	0.12	0.00	0.02	0.08	0.75	0.00	0.00	0.99
SY550	Ohta (1989) ^a	1.28	15.25	0.26	2.66	22.73 ^a	29.44	–	–	28.91	77.80	0.03	0.26	0.00	0.03	0.22	0.49	0.00	0.00	0.98
RCD379-1	Sahlström, Arribas, et al. (2017), Sahlström, Blake, et al. (2017)	0.03	14.55	0.00	–	19.59	35.79	1.05	0.00	29.46	100.47	0.00	0.24	0.00	0.00	0.18	0.57	0.02	0.00	0.95
ZI-41	Semenyak et al. (1994) ^a		11.71	1.55	0.46	24.68 ^a	31.47		0.25	28.71	74.15	0.000	0.205	0.016	0.004	0.239	0.536	0.000	0.002	0.997
5497-3-28	Sinclair et al. (2006)	10.3	3.38	0.05	0.06	5.59	47.71		0.22	32–42	67.31	0.18	0.05	0.00	0.00	0.05	0.72	0.00	0.00	1.00
17	Shimizu and Morishita (2012)	4.81	3.04	0.06	0.03	5.48	52.37		0.44	32.44	98.67	0.086	0.048	0.001	0.000	0.048	0.801	0.000	0.004	1.012
3	Tsushima et al. (1999) ^a	6.98	13.06	0.33	3.7	16.3 ^a	28.59			30.12	82.78	0.133	0.218	0.003	0.033	0.151	0.464	0.000	0.000	0.998
7	Voudouris et al. (2022)	0.00	7.00		0.00	6.06	55.27	0.00	0.00	31.58	99.91	0.000	0.111	0.000	0.000	0.053	0.850	0.000	0.000	0.989
EK26196.5	Willson (2019) ^a	11.20	11.79	0.00	0.13	19.77 ^a	24.88		0.69	30.04	78.73	0.213	0.197	0.000	0.001	0.183	0.404	0.000	0.007	0.995

^aAnalyses identified by the respective authors as Zn-In-minerals, differentiating them from In-rich sphalerite.

chalcophile elements (e.g., Heinrich, 2005; Márquez-Zavalía & Heinrich, 2016; Sillitoe, 1973, 1996; and references therein).

The In-Cu-rich sphalerite is chemically similar to roquesite CuInS_2 (Picot & Pierrot, 1963). However, roquesite structure crystallizes in a chalcopyrite-type, has no Zn and displays a structural ordering between Cu and In, while in this In-Cu-rich sphalerite all metals (Zn, Cu, and In) occupy a single crystallographic site (Figure 8), randomly distributed at the 4a position (structural disorder). Sakuraiite $(\text{Cu, Zn, Fe})_3(\text{In, Sn})\text{S}_4$ is the most related mineral; nevertheless, whereas sakuraiite shows $P432$, $P-43m$, $Pm3m$ symmetry (Kissin & Owens, 1986), this In-Cu-rich sphalerite adopts the $F-43m$ symmetry, and contrary to sakuraiite it does not contain Sn as essential or minor chemical element. Structurally related ishiharaite $(\text{Cu, Ga, Fe, In, Zn})\text{S}$ (Márquez-Zavalía et al., 2014) contains small quantities of Zn and a significant amount of Ga, which is absent in the In-Cu-rich sphalerite. The recently described richardsite, $\text{Zn}_2\text{CuGaS}_4$ (Bindi & Jaszczak, 2020) is also chemically similar to the In-Cu-rich sphalerite, with Ga instead of In, but it is tetragonal, being a new member of the stannite group with space group $I-42m$ (Figure 9).

The minerals described by Ohta (1980, 1989) and Semenyak et al. (1994) as a “Zn-In mineral” phase, have a similar chemical composition (Table 6), but some of the physical properties of the In-Cu-rich sphalerite are slightly different. Murakami and Ishihara (2013) described an undetermined Zn–In mineral in Huari Huari (Bolivia), with an average Fe content of 12.79 wt%, which is different from the variety of sphalerite described here, characterized by an average content of 0.06 wt% Fe (Table 6, Figure 9).

In agreement with Frenzel et al. (2016, 2019), we conclude that, due to the strong correlation between Cu and In in the structure of sphalerite, the Cu-rich Pb–Zn systems are more prospective for In than the ones poor in Cu. Although Capillitas is a polymetallic ore deposit that has been mined for Cu, Pb, Au and Ag since the mid-19th century, due to repeated ore beneficiation problems in the last 50 years it has been mined almost exclusively for rhodochrosite. Therefore, it would be interesting to check the potential of the dumps and the nonmined areas of the different veins, especially those from Restauradora, Nueva Esperanza and La Rosario, for this critical element, as well as Ga and Ge also present in them (Bindi et al., 2017; Márquez Zavalía, 2005; Márquez-Zavalía et al., 2014, 2020; Paar et al., 2004; Putz et al., 2006).

When the study was completed, and regardless of the differences in optical properties and chemical composition, since Zn remains the main single cation, despite the large substitution of Cu + In in the structure, the mineral is still classified as sphalerite, being

the richest In-Cu sphalerite described in the literature (Tables 3 and 6, Figure 9).

6 | CONCLUSIONS

Among all the veins in Capillitas, Restauradora is the one that has the In-Cu-richest sphalerite.

The In-Cu-rich sphalerite looks like homogeneous regular sphalerite under the polarizing microscope when the In concentrations in the sphalerite structure are lower than 22 wt% In (about 0.20 *apfu*). When these values are exceeded, the ribbons of In-Cu-rich sphalerite become visible.

The In-Cu-rich sphalerite described here is the In-richest variety described in literature.

The porphyry intrusions associated with the Farallón Negro Volcanic Complex may have been the source of the low- to intermediate-salinity fluid that carried the indium and other chalcophile elements.

Exploration for indium in Capillitas should be encouraged, especially in dumps and shafts of the Restauradora, Nueva Esperanza and La Rosario veins were the presence of In, Ga, and Ge was recognized.

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CONFLICT OF INTEREST STATEMENT

Dr. Yasushi Watanabe is the Editor-in-Chief of the journal and the co-author of this article. He was excluded from the peer-review process and all editorial decisions related to the acceptance and publication of this article. Peer-review was handled independently by RGE Journal editorial office and Dr. Tatsuo Nozaki as the Editor to minimize bias. The authors declare that they have no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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